Facial Expression Recognition with Deep Learning and Attention Mechanisms: A Systematic Review

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***Abstract*—This comprehensive literature review meticulously examines the dynamic field of Facial Expression Recognition (FER), particularly highlighting the transformative role of deep learning and attention mechanisms in advancing the state of the art. By meticulously analyzing 53 seminal papers, we highlight the dominant roles of CNN, ResNet, and LSTM, with particular attention to the Region Attention Network (RAN). These technologies have significantly advanced the accuracy and efficiency of FER systems, marking a shift towards more sophisticated analysis methods. Our review not only synthesizes the current state of FER advancements but also identifies key trends, challenges, and ethical considerations shaping future research directions. We emphasize the need for culturally sensitive, robust FER systems and the potential of multimodal approaches to enrich the understanding and interpretation of facial expressions. This review aims to provide a foundational resource for future studies, guiding the development of more nuanced, effective, and ethical FER technologies, with a special note on its emerging applications in monitoring and enhancing employee emotional well-being in Human Resources (HR).**

Keywords— Facial Expression Recognition, Deep Learning, CNN, Attention Mechanisms, Systematic Literature Review

I. Introduction

FER stands at the forefront of computational behavioral analysis, revolutionizing domains like robotics, digital marketing, and human-computer interaction. Integrating deep learning and attention mechanisms into FER systems has opened new avenues for understanding and interpreting human emotions. This systematic literature review critically examines these technological advancements, focusing on their accuracy and efficiency in diverse application scenarios [1],[27],[53].

FER has gained unprecedented importance due to its ability to decipher emotional cues integral to human interaction. Robots enable machines to understand and respond to human emotions, while digital marketing allows for tailoring strategies based on consumer emotional responses. These advancements in FER are not only enhancing user experience but are also pivotal in developing intelligent systems capable of empathetic responses [2],[16],[4].

Replicating the human ability to interpret facial emotions in computational systems presents a notable challenge. This task requires the algorithmic mimicry of intricate human perception capabilities, accounting for the extensive variability in facial expressions. The complexity is further compounded by cultural and individual differences, which must be accurately captured and interpreted by FER systems [8],[14],[17].

Beyond its application in robotics, digital marketing, and human-computer interaction, the utility of FER, particularly when enhanced with deep learning and attention mechanisms, notably expands into human resources (HR). Within HR, applying FER technology is a pivotal tool for deciphering and managing employees' emotional states. This aspect is essential in fostering a workplace environment that is both dynamic and supportive. By interpreting facial expressions, HR professionals can better understand factors like job satisfaction, stress levels, and general well-being among employees. This application becomes especially relevant in scenarios where understanding the emotional undercurrents of the workforce is key to enhancing overall workplace harmony and productivity [44],[55],[58].

This review aims to address crucial questions surrounding the integration of deep learning and attention mechanisms in FER. It explores the accuracy of these methods in recognizing a wide range of emotions, their effectiveness across different datasets, and their applicability in various real-world scenarios. Additionally, the review investigates the future potential of these technologies in advancing the field of emotion recognition from facial images [6],[29],[31].

Fig. 1 in this review delineates a quantitative analysis of journal sources, showcasing a comparison of publications focused on Facial Emotion Recognition against those emphasizing FER with Deep Learning and Attention Mechanisms. Sourced from reputed journals and databases like MDPI, Springer, Hindawi, IEEE Xplore, and Science Direct, this analysis spans research from 2014 onwards.

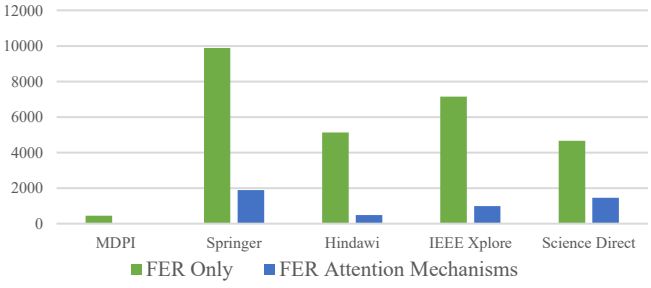


Fig. 1. Quantitative Analysis Of Journal Sources

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The graphical representation in Fig. 1 quantifies the research output in these areas and highlights the shifting focus within the academic community. It underscores a growing interest in exploring the synergies between deep learning and attention mechanisms for more accurate and context-aware emotion recognition. This trend reflects the field's dynamic progression and its expanding scope, encompassing various practical applications, including those in HR for emotion detection and management among employees [48],[56],[59].

In the final analysis, this scholarly review aspires to meticulously delineate the current research expanse in the FER domain. It aims to elucidate seminal breakthroughs and prognosticate trajectories for incipient scholarly inquiry. This endeavor is envisaged to endow both the academic milieu and the industrial echelons with incisive, perceptive elucidations, thereby fomenting the advent of groundbreaking methodological paradigms and pragmatic applications in the FER sphere.

II. Methodology

In this systematic review, our investigative focus centers on integrating deep learning and attention mechanisms within FER. This comprehensive evaluation of the existing literature is carefully constructed to delve deeply into this domain's myriad advancements and challenges. We intend to thoroughly understand how these technological integrations, which represent the cutting edge of computational methods, collectively contribute to enhancing the accuracy and efficiency of FER systems [1],[27].

A. Review Method

This systematic review employs the latest Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 guidelines to meticulously evaluate the integration of deep learning and attention mechanisms in FER [58]. The methodology includes a thorough examination of existing literature, focusing on studies that implement these advanced computational techniques. The review process involved identifying relevant studies, assessing their methodological rigor, and synthesizing their findings to understand the impact of these technologies on the accuracy and efficiency of FER systems.

B. Research Questions

This review seeks to address the following questions:

- **Deep Learning Algorithms:** Which algorithms are most used in FER?
- **Integration of Attention Mechanisms:** How are these mechanisms incorporated into FER systems?
- **Accuracy Improvements:** What advancements in accuracy are observed with these integrations?

C. Search strategy

Our search strategy involved extensive queries across critical academic databases, including MDPI, Springer, Hindawi, IEEE Xplore, and Science Direct. Keywords such as "Facial Expression Recognition," "Deep Learning," and "Attention Mechanisms" were selected to capture a comprehensive range of relevant studies, mainly from publications dating from 2014 onwards [48],[56],[59].

The inclusion criteria were stringent, focusing on studies explicitly exploring integrating deep learning with attention mechanisms in FER. In contrast, the exclusion criteria filtered out articles that did not directly align with our research questions, were not in English or needed a focused exploration of technological integration [3],[16]. From an initial pool of 210 articles, 53 were identified as most pertinent to our study's objectives.

D. Selection and Data Collection Process

Fig. 2 illustrates the systematic review process, beginning with an initial search of 210 articles and followed by title and abstract screening. Relevant articles proceed to a full-text review, while irrelevant ones are excluded. The 53 selected articles undergo detailed methodological analysis and quality assessment. The final step involves data synthesis, summarizing the findings on integrating deep learning and attention mechanisms in FER.

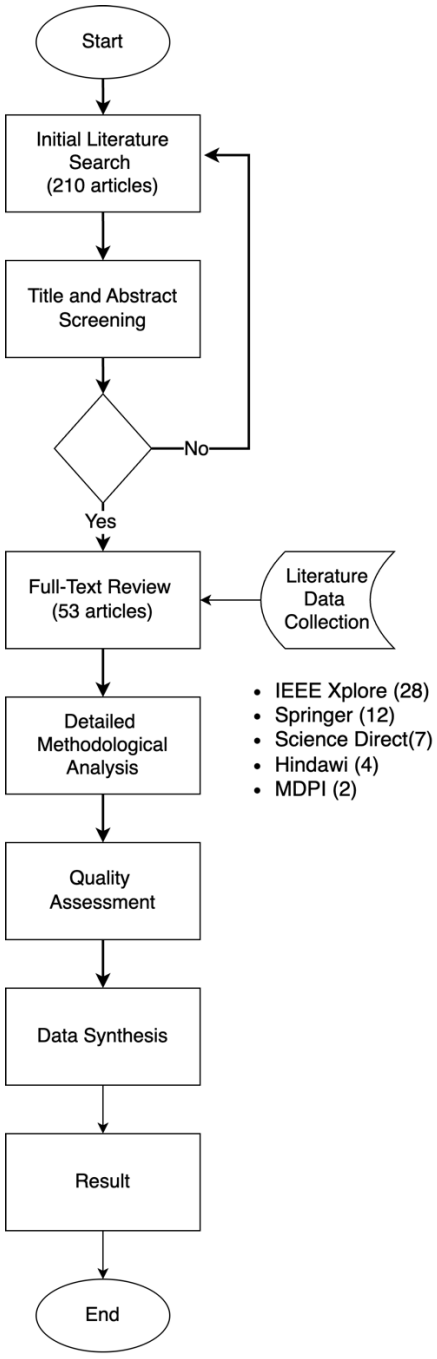


Fig. 2. Systematic Review Process

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E. Synthesis Methods

This analysis entailed cataloguing the types of technologies used and critically evaluating their effectiveness in enhancing FER accuracy. The synthesis of these findings sought to identify common trends, innovative approaches, and divergences within the research landscape. A key aspect of our methodology was quality assessment. Each article was evaluated for methodological robustness, journal reputation, and peer-review status to ensure that our review was grounded in high-quality, reliable research findings. The outcomes of this literature review are anticipated to yield pivotal insights into the progress and effectiveness of combining deep learning and attention mechanisms within FER, influencing the trajectory of future research in this area and potentially catalyzing the development of FER systems that are both more precise and efficient.

III. Results

A. Deep Learning Approaches

Deep learning has significantly influenced the field of FER, with techniques like Convolutional Neural Networks (CNNs), Residual Networks (ResNet), and Long Short-Term Memory networks (LSTM) playing pivotal roles. These algorithms have been instrumental in enhancing the recognition and analysis of facial expressions, as evidenced in the literature [6] ,[9] ,[13] ,[14] ,[22],[23],[26],[35],[36],[42] ,[53] for CNN; [9],[14],[27],[33] for ResNet; [21],[35],[48] for LSTM]. An analysis of 53 papers revealed that three main algorithms CNN, ResNet, and LSTM dominated the field.

CNNs are particularly noted for their efficacy in processing grid-like topology data, such as images, making them ideal for facial feature extraction and recognition tasks. ResNet's unique residual learning framework aids in training deeper networks and LSTMs are known for their ability to handle sequential data, adding depth to the analysis in FER studies. Fig. 3 illustrates the distribution of deep learning algorithms in FER research, showing the dominance of CNN (72.73%), followed by ResNet (18.18%) and LSTM (9.09%) in the analyzed papers.

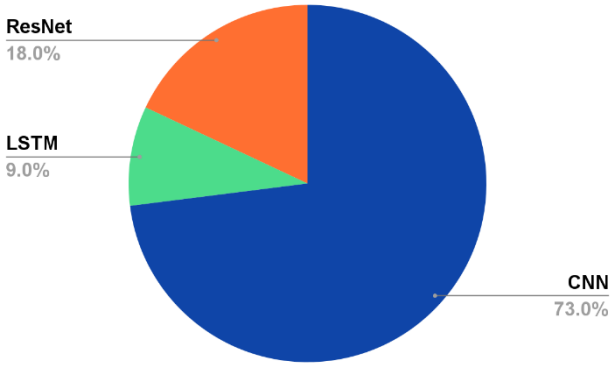


Fig. 3. This chart illustrates the dominance of CNN (72.73%), followed by ResNet (18.18%) and LSTM (9.09%) in the analyzed papers

B. Attention Mechanisms

The role of attention mechanisms in augmenting deep learning models for FERhas become a cornerstone in recent advancements within the field. By homing in on the essential features within facial expressions, these mechanisms have significantly improved both the accuracy and efficiency of recognition systems. This trend underscores the shift towards more nuanced and sophisticated approaches in FER, reflecting the field's ongoing evolution. This trend underscores the shift towards more nuanced and sophisticated approaches in FER, reflecting the field's ongoing evolution. An exhaustive analysis of the relevant literature has identified the three most frequently cited attention mechanisms instrumental in this progression. These mechanisms are praised for their accuracy and efficiency in emphasizing key features in facial expressions, leading to more refined recognition capabilities.

One such mechanism that stands out is the Region Attention Network (RAN). The RAN is particularly noteworthy for its ability to enhance the focus on pertinent facial areas, thereby sharpening expression recognition accuracy. This mechanism's proficiency lies in its capacity to aggregate and embed region-specific features and in its application of attention weights that prioritize significant regions within facial images [8],[54]. These attributes are instrumental in achieving a more granular level of recognition, setting a high standard in the field.

TABLE I. Attention Mechanisms in FER

Attention Mechanism	Subhead	Citations
Attention Module in MobileNet	Introduces an attention module into MobileNet, significantly improving feature extraction for FER.	[40],[43]
RAN with Attention Weights	Utilizes attention weights in RAN to prioritize significant regions in facial images.	[8],[54]
Network Enhancing Spatial Correlations	Enhances spatial correlations between facial regions using a cross-modality attention mechanism.	[26],[53]

Attention modules in architectures like MobileNet [40],[43] and networks that enhance spatial correlations [26],[53] demonstrate the integration of attention mechanisms to improve feature extraction capabilities. This trend towards more nuanced and sophisticated approaches in FER signifies significant progress in the field.

As illustrated in Fig. 4 and discussed in the referenced literature, the attention mechanism in FER plays a critical role in enhancing the accuracy and effectiveness of expression recognition systems. This mechanism, similar to the human visual system's selective focus, allows neural networks to concentrate on the most relevant features of facial expressions, such as the eyes and mouth, which are crucial for accurate expression recognition [27],[44].

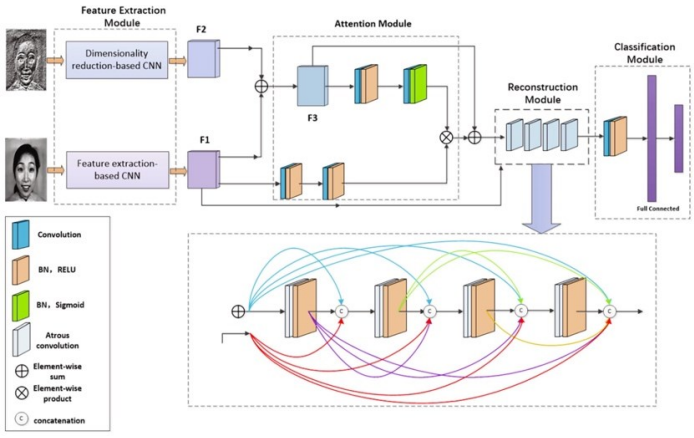


Fig. 4. How attention mechanisms are applied in the context of FER [6].

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In the process outlined in Fig. 4, the attention mechanism typically involves two key components. The first is the trunk branch, which extracts the primary feature set from the facial image. This branch serves as the backbone of the attention system, processing the image to extract a comprehensive set of features[27],[44]. The mask branch's second component is integrated with Local Binary Patterns (LBP) features to generate attention maps. These maps highlight the most pertinent information for expression recognition, with the LBP features aiding in capturing the textural and micro-pattern details essential for identifying subtle expressions[27],[44].

The attention mechanism refines the feature set by applying an element-wise product between the primary features (F_p) and the attention maps (F_m). This operation results in refined feature maps (F_{refine}), where the network's focus is directed toward the most relevant features for expression recognition. The refinement process is governed by:

$$F_{refine}=F_P \text{ x } F_m \quad (1)$$

The mechanism refines the feature set by applying an element-wise product between these primary features and the attention maps, resulting in a focused and refined feature set. This refined set is more attuned to the relevant features necessary for accurate expression recognition, demonstrating the mechanism's ability to enhance classification performance by prioritizing essential elements while suppressing unimportant ones[27],[44]. The attention heatmaps, part of this process, visually represent these areas of focus, indicating the areas containing the most useful features for recognizing different expressions[27],[44].

C. FER with Attention Mechanism

The exploration of advanced FER systems has been at the forefront of significant scholarly inquiry, signifying a convergence of state-of-the-art computational techniques and deep analytical insights. This literature review is an extensive examination of 53 critical papers, offering a broad perspective on the dominant strategies and innovations within the scope of FER. In examining these comprehensive studies, our objective is to illuminate the critical contributions of deep learning algorithms and attention mechanisms in enhancing the precision and effectiveness of FER systems. This analysis, deeply rooted in empirical evidence, provides a novel interpretation of how these technological advancements reshape the landscape of facial expression analysis. Our analysis is firmly rooted in the diverse data encapsulated within these papers, ensuring a comprehensive and accurate portrayal of the field's current trajectory and prospective evolution [27].

Our analysis significantly focuses on three primary deep learning algorithms within FER : Convolutional Neural Networks (CNNs), Residual Networks (ResNet), and LSTM. CNNs are particularly noted for their effectiveness in processing images, a type of grid-like topology data, and are prominently featured in a vast array of studies [6],[9],[13],[14],[22],[23],[26],[35],[36],[42],[53]. They are widely acclaimed for their ability to efficiently extract and recognize facial features, thus becoming fundamental to FER research. ResNet, known for its innovative residual learning framework, aids in training substantially deeper networks and is acknowledged in 18.18% of the analyzed literature [9],[14],[27],[33]. Meanwhile, LSTMs, recognized for their adeptness in managing sequential data, add a deeper layer of analysis to facial expressions and are utilized in 9.09% of the studies [21],[35],[48].

Attention mechanisms have fundamentally transformed FER systems by emphasizing essential features, significantly enhancing accuracy and efficiency. A prime example is the Region Attention Network (RAN), which has demonstrated remarkable efficacy in improving expression recognition accuracy by focusing on pertinent facial regions [8],[54]. RAN works by aggregating and embedding region-specific features and applying attention weights to prioritize significant areas within facial images. Additionally, integrating attention modules in architectures like MobileNet has substantially improved feature extraction for FER [40],[43]. At the same time, other networks have successfully enhanced spatial correlations between facial regions, further refining recognition capabilities [26],[53]. These advancements indicate the field's shift towards more nuanced and sophisticated FER approaches. Overview of Attention Mechanisms and Their Impact on FER Accuracy.

Given the profound impact of attention mechanisms on FER accuracy and efficiency, Table II. has been constructed to compare the various strategies employed and their respective results. By strategically concentrating on key facial features, these mechanisms have significantly advanced the precision and effectiveness of FER systems.

TABLE II. OVERVIEW OF ATTENTION MECHANISMS AND THEIR IMPACT ON FER ACCURACY

Database	Attention Mechanisms	Accuracy	Citation
- FER2013 - CK+ - JAFFE	Attention mechanism is used to adaptively emphasize important information and suppress irrelevant information.	74%	[35]
- AFEW - CK+ - Oulu - CASIA	Spatial features are input to the hybrid attention module.	98%	[30]
- CK+ - JAFFE - KDEF - FER-2013	The proposed attention network consists of three modules: global representation, channel representation, and dimension representation.	98%	[56]
- RAF	Attention mechanism mimics human selective attention in pattern recognition.	85%	[57]
- KDEF - RAF-DB - FERG-DB	The proposed lightweight attention DCNN (LA-Net) uses squeeze-and-excitation (SE) modules.	52%	[36]

Integrating deep learning and attention mechanisms has led to significant advancements in FER accuracy. For example, the Region Attention Network (RAN) has been particularly influential, enhancing the focus on pertinent facial regions and significantly sharpening expression recognition accuracy. Studies have evidenced that the RAN

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and region-biased loss notably improve FER performance [8],[54]. Furthermore, the Vision Transformer with Attentive Pooling for FER in the Wild (APViT) achieves a high accuracy rate of 97.30% in FER tasks, demonstrating the effectiveness of these innovative approaches [6].

Moreover, attention-based methods have yielded high accuracy rates across various datasets. For instance, studies have noted improvements in dimensional emotion estimation, describing a broader spectrum of human emotions and contributing to a more nuanced understanding and classification of facial expressions [7]. In particular, datasets such as FER2013 and CK+ have seen improved performance due to these attention mechanisms, reflecting the systems' enhanced capability to focus on the most relevant features for expression recognition, thereby boosting overall accuracy [29]. Applying an element-wise product between primary features and attention maps refines the feature set. This technique directs the system's focus toward the most pertinent features for expression recognition, enhancing classification performance and leading to more accurate and reliable FER systems [27],[44]. Similarly, the proposed lightweight attention DCNN (LA-Net), which uses squeeze-and-excitation (SE) modules, focuses on salient features to improve recognition rates in varied datasets like KDEF, RAF-DB, and FERG-DB [36].

IV. Conclusions and Future Work

This review has explored FER's development, highlighting the role of deep learning and attention mechanisms in enhancing emotional recognition. Techniques like CNN, ResNet, and LSTM have been pivotal in interpreting facial expressions, while attention mechanisms, especially the Region Attention Network (RAN), offer focused feature analysis. These advancements have expanded FER's applicability in various fields. The research is shifting towards creating robust, adaptable FER systems, integrating diverse data to improve accuracy and reduce biases. With the progression of FER, particularly in sensitive sectors like HR, balancing technological innovation with ethical considerations of privacy and emotional integrity is essential for its responsible use.

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